EXPERIMENT 4 - Impulse Response, Causality, and Stability Analysis of LTI Systems

INPUT CODE

```
clc;
clear;
close all;

%% EXPERIMENT 4 - TASK B
% Anti-causal Stable CT and Two-sided Stable DT
%
% CT: h(t) = exp(t)u(-t)
% DT: h[n] = (0.7)^|n|

%% ============================================================
% 1. CONTINUOUS-TIME IMPULSE RESPONSE
%    h(t) = exp(t)u(-t)
% ============================================================

% Continuous-time vector and sampling interval
dt = 0.01;
t = -5:dt:5;

% Unit step u(-t)
u_minus_t = double(t <= 0);

% CT impulse response
h_ct = exp(t) .* u_minus_t;

% Plot CT impulse response
figure;
plot(t, h_ct, 'LineWidth', 2);
grid on;
xlabel('Time t');
ylabel('h(t)');
title('CT Impulse Response: h(t) = e^t u(-t)');
%% Finite-window absolute area

area_ct = trapz(t, abs(h_ct));

fprintf('CT finite-window absolute area = %.4f\n', area_ct);


%% ============================================================
% 2. DISCRETE-TIME IMPULSE RESPONSE
%    h[n] = (0.7)^|n|
% ============================================================

% Discrete-time index vector
n = -20:20;

% DT impulse response
h_dt = (0.7).^abs(n);

% Plot DT impulse response
figure;
stem(n, h_dt, 'filled');
```

```
grid on;
xlabel('Index n');
ylabel('h[n]');
title('DT Impulse Response: h[n] = (0.7)^{|n|}');

%% Finite-window absolute sum
sum_dt = sum(abs(h_dt));

fprintf('DT finite-window absolute sum = %.4f\n', sum_dt);


%% ============================================================
% 3. CAUSALITY OBSERVATION
% ============================================================

% CT: Check negative-time values
if all(h_ct(t < 0) == 0)
   ct_causal = 'Causal';
else
   ct_causal = 'Anti-causal';
end

% DT: Check negative-index values
if all(h_dt(n < 0) == 0)
   dt_causal = 'Causal';
else
   dt_causal = 'Non-causal (Two-sided)';
end


%% ============================================================
% 4. BIBO STABILITY
% ============================================================

% CT analytical integral:
% Integral from -infinity to 0 of exp(t) dt = 1

ct_integral_theoretical = 1;

% DT analytical infinite sum:
% Sum from n=-infinity to infinity of (0.7)^|n|
%
% = 1 + 2*(0.7 + 0.7^2 + ...)
% = 1 + 2*(0.7/(1-0.7))
% = 17/3 = 5.6667

dt_sum_theoretical = 1 + 2*(0.7/(1-0.7));


%% ============================================================
% 5. DISPLAY FINAL RESULTS
% ============================================================

fprintf('\n==========================================\n');
fprintf('      TASK B RESULTS\n');
fprintf('==========================================\n');
```

```
fprintf('\nCT System: h(t) = exp(t)u(-t)\n');
fprintf('Causality      : %s\n', ct_causal);
fprintf('Finite-window absolute area = %.4f\n', area_ct);
fprintf('Theoretical absolute integral = %.4f\n', ct_integral_theoretical);
fprintf('BIBO Stability  : Stable\n');

fprintf('\nDT System: h[n] = (0.7)^|n|\n');
fprintf('Causality      : %s\n', dt_causal);
fprintf('Finite-window absolute sum = %.4f\n', sum_dt);
fprintf('Theoretical infinite absolute sum = %.4f\n', dt_sum_theoretical);
fprintf('BIBO Stability  : Stable\n');

fprintf('\n==========================================\n');
```

OUTPUT



